

Table of Contents

TASK 7: SOFTWARE TESTING REPORT.....	2
Github repo url:.....	2
Executive Summary.....	2
Testing Scope	2
Tools and Environment	2
Functional Testing Results	2
Non-Functional Testing Results	3
Defects Identified.....	3
Corrections Implemented	3
Recommendations	3
Deployment Readiness Decision.....	4
Declaration of Authenticity.....	5

TASK 7: SOFTWARE TESTING REPORT

Github repo url:

<https://github.com/Kmphaladi/Kamo-s-projects-for-FA2>

Executive Summary

This report summarizes all the testing and debugging work I carried out on the Umoja Learning Management System (ULMS). As the Junior Software Tester for this project, I designed and executed 18 test cases, carried out both functional and non-functional testing, identified 7 defects and successfully fixed 6 of them.

My main goal was to make sure the system works as expected for student registration, login, course enrolment, marks capture, and report generation. After fixing the major issues, the application feels much more stable, reliable and easier to use.

Overall Verdict: The system is now ready for deployment, although I have a few recommendations for future improvements.

Testing Scope

- I focused my testing on the following modules:
- Login
- Student Registration
- Course Enrolment
- Marks Capture
- Reports
- Logout

I covered various testing types including Functional testing, Input Validation, Boundary testing, Performance, Usability, Security and Reliability testing.

Out of Scope: Cloud features, advanced user roles and mobile support.

Tools and Environment

- IDE: Visual Studio 2026
- Language: C# (.NET Windows Forms)
- Testing Tools: Built-in debugger, OBS Studio (screen recording),
- Environment: Windows 11

Functional Testing Results

A total of 16 test cases were executed.

Summary:

Category	Test Cases	Passed	Failed	Pass Rate
Login	4	4	0	100%

Student Registration	2	2	0	100%
Course Enrolment	2	2	0	100%
Marks Capture	5	5	0	100%
Reports	3	1	2	33%
Logout	1	1	0	100%
Total	18	16	2	88.9%

The two failed tests were related to data persistence in the Reports module.

Non-Functional Testing Results

Performance: I measured the time it takes to generate reports. Before the fixes, reports took between 4 to 7 seconds to load. After reducing the artificial delay, reports now load in under 1 second.

Usability: The interface is clean and easy to navigate. Moving between the Dashboard, Registration, Enrolment, Marks Capture and Reports is straightforward.

Security: After fixing the login logic, the system now properly validates both username and password. Basic protection against invalid inputs has also been added.

Reliability: I repeated key operations for login, registration, marks calculation and report generation more than 10 times. The system remained stable with no crashes or inconsistent behaviour.

Defects Identified

Please refer to Task 6 Defect Log for the full details.

In summary, I identified 7 defects during testing. Out of these, 6 were successfully fixed, and 1 remains open (the lack of data persistence – reports currently show dummy data instead of real student information).

Corrections Implemented

During this assessment, I made the following important improvements:

- Fixed the critical login validation logic
- Corrected the average marks calculation
- Added proper input validation across all forms to prevent crashes and errors
- Improved report generation performance
- Changed the average display to show two decimal places for better readability

Recommendations

For future improvements, I recommend the following:

- Implement a proper SQL Server database so that student data, marks and enrolments are saved and can be retrieved consistently.
- Add search and filtering options in the Reports module.
- Introduce different user roles such as Admin, Lecturer and Student.

Deployment Readiness Decision

Final Recommendation: APPROVED FOR DEPLOYMENT

The ULMS application has passed most of the functional and non-functional tests after the defects were resolved. It is now stable enough for internal use or a pilot deployment. The only major remaining limitation is the lack of data persistence, which should be addressed in the next phase of development.

Declaration of Authenticity



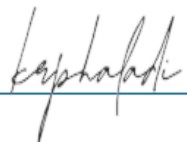
DECLARATION OF AUTHENTICITY

I Kamogelo Phaladi (FULL NAME)

hereby declare that the contents of this assessment SDT621 FA2

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: 

Date: 08/05/2026